COGNIFYZ IT SOLUTIONS

Machine Learning Internship

TASK 1

Restaurant Rating Prediction

Predicting Aggregate Restaurant Ratings Using Regression Models

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Internship Domain	Machine Learning
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Dataset	Restaurant Dataset (Dataset.csv)

Based on the completed Task 1 notebook and Cognifyz Machine Learning task requirements.

1. Objective

The objective of this task is to build a machine learning regression model that can predict the **Aggregate Rating** of a restaurant using available restaurant-related features. The workflow includes handling missing values, removing unsuitable or target-derived fields, encoding categorical variables, splitting the data into training and testing sets, training regression models, evaluating them using Mean Squared Error (MSE) and R^2 score, tuning the decision tree, and interpreting influential features.

2. Task Requirements and Scope

The Cognifyz Task 1 specification requires the dataset to be preprocessed, categorical variables to be encoded, the data to be divided into training and testing sets, regression models to be trained, model performance to be evaluated using metrics such as MSE and R-squared, and the influential features affecting restaurant ratings to be analyzed. The implementation in this report follows those requirements.

Requirement	Implementation in this task
Preprocessing	Handled missing Cuisines values and removed unsuitable / target-derived features.
Encoding	One-hot encoded selected categorical features.
Train/Test Split	80:20 split with random_state=42.
Regression Models	Linear Regression and Decision Tree Regressor.
Evaluation	MSE and R^2 on the test set.
Interpretation	Compared models, analyzed overfitting, tuning, and feature importance.

3. Dataset Description

The dataset used for this task contains **9,551 restaurant records and 21 original columns**. The target variable is **Aggregate rating**. The data includes restaurant identity, location, cuisine, pricing, service options, customer votes, and rating-related information.

Observed Dataset Details

Dataset Item	Observed Value
Rows	9,551
Original columns	21
Missing values in Cuisines before handling	9
Train/Test split	7,640 / 1,911
Processed feature count	165

4. Data Loading and Exploration

The dataset was loaded into a pandas DataFrame and its structure was inspected using shape, columns, information summaries, missing-value checks, and selected exploratory operations. This helped identify feature types, missing values, redundant fields, and target-derived attributes before modelling.

Only the **Cuisines** column contained missing values, with 9 missing records. These values were replaced with the label **Unknown** so that the restaurant records were retained while avoiding missing entries in that feature.

5. Data Preprocessing

5.1 Missing-Value Handling

The 9 missing values in Cuisines were replaced with Unknown. No restaurant records were removed for this missing-value issue.

5.2 Removing Irrelevant and Target-Leaking Features

Restaurant ID was removed because it is an identifier rather than a meaningful predictive feature. Switch to order menu was removed because it contained a constant value. Rating text and Rating color were removed because they directly correspond to the Aggregate rating and could introduce target leakage.

5.3 Removing High-Cardinality Features

Restaurant Name, Address, Locality, Locality Verbose, and Cuisines were removed from the final modelling feature set because these fields have very high cardinality. One-hot encoding them directly would create a large sparse feature space and make the model unnecessarily complex.

5.4 Feature Encoding

The selected categorical variables - Country Code, City, Currency, Has Table booking, Has Online delivery, and Is delivering now - were encoded using OneHotEncoder with handle_unknown='ignore'. Numerical variables such as Longitude, Latitude, Average Cost for two, Price range, and Votes were passed through without scaling. A ColumnTransformer was used to apply the appropriate preprocessing to each feature group. The resulting representation contained **165 processed features**.

6. Train-Test Split

The processed dataset was divided using an 80:20 train-test split with random_state=42. This produced **7,640 training records** and **1,911 testing records**. The preprocessing transformer was fitted only on the training data and then applied to the test data to prevent information from the test set from influencing preprocessing.

7. Model Building

7.1 Linear Regression

Linear Regression was used as the initial baseline because Aggregate rating is a continuous target. The model was trained on the preprocessed training data and evaluated on the unseen test set.

7.2 Decision Tree Regression

A Decision Tree Regressor was then trained to capture non-linear relationships between the restaurant features and Aggregate rating. The initial model used random_state=42.

8. Model Evaluation

Model	Mean Squared Error	R^2 Score
Linear Regression	1.5216	0.3315
Decision Tree Regressor	0.1988	0.9127

The initial Decision Tree Regressor clearly outperformed Linear Regression on the test set, achieving a higher R^2 score and a much lower MSE. This indicates that the tree captured the available relationships more effectively.

8.1 Overfitting Analysis

The initial Decision Tree achieved a training MSE of approximately 0.00001832 and a training R^2 score of approximately 1.0000, while its test performance was weaker. The large difference between training and testing performance indicated overfitting. The model complexity was therefore reduced using the max_depth parameter.

9. Model Tuning

Max Depth	MSE	R^2 Score
5	0.1066	0.9532
10	0.1117	0.9509
15	0.1496	0.9343

Among the tested depths, **max_depth=5** provided the best test performance, with the lowest MSE and highest R^2. The shallower tree generalized better than the deeper alternatives.

10. Final Model Selection

The final model was selected as a Decision Tree Regressor with **max_depth=5** and **random_state=42**. It provided the strongest test performance among the tested tree depths while reducing the overfitting observed in the unrestricted tree.

11. Feature Importance Analysis

Feature importance was examined using the feature_importances_ attribute of the final Decision Tree. **Votes** was by far the most influential feature, followed by **Longitude** and **Latitude**. Other features had substantially smaller contributions.

The high importance of Votes indicates a strong association between customer voting activity and the model prediction of Aggregate rating. Feature importance is model-specific and should be interpreted as predictive contribution rather than proof of direct causation.

12. Actual vs Predicted Visualization

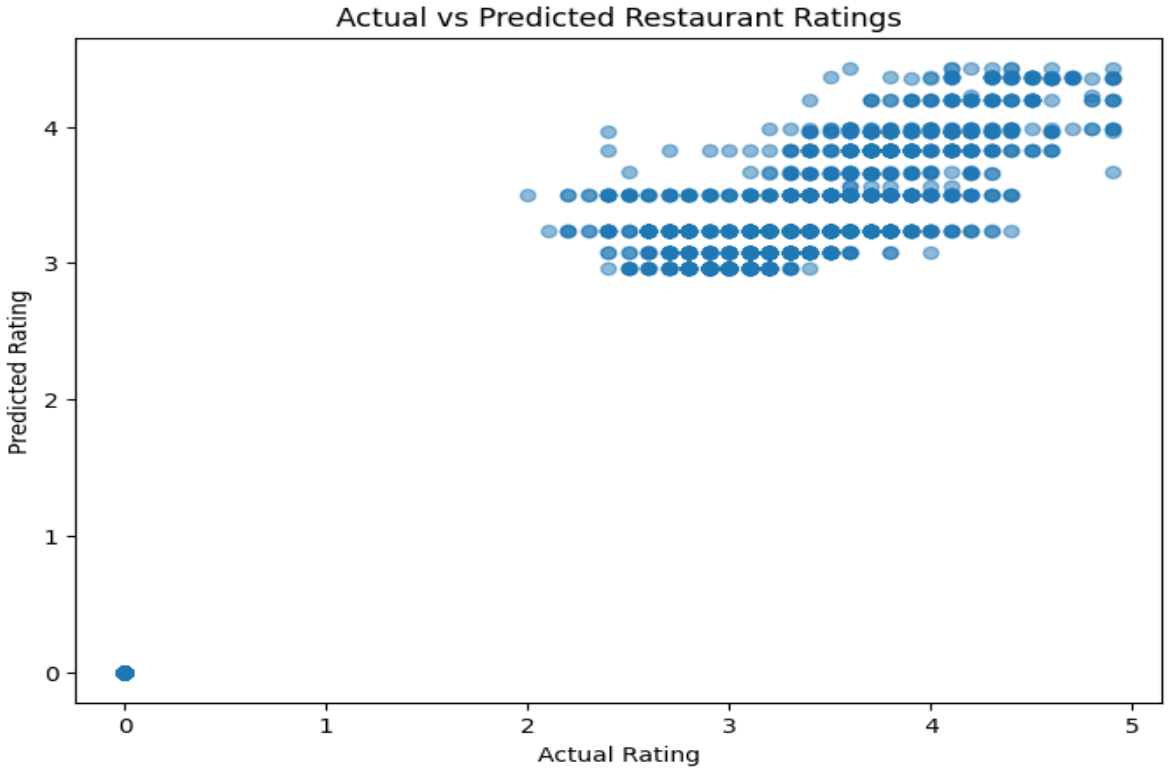


Figure 1: Actual vs Predicted Aggregate Rating

The scatter plot compares actual Aggregate rating values from the test set with predictions produced by the final Decision Tree. Points closer to the ideal diagonal relationship indicate better agreement between actual and predicted values.

13. Results and Interpretation

The modelling results show that Decision Tree Regression performed substantially better than Linear Regression for this restaurant rating prediction task. The initial Decision Tree exhibited clear signs of overfitting, so tuning was performed by restricting the maximum depth. The tuned model with max_depth=5 achieved a test R^2 score of approximately **0.9532** and an MSE of approximately **0.1066**.

The feature importance analysis identified Votes as the dominant feature used by the final Decision Tree, while geographic features such as Longitude and Latitude had smaller contributions. These results provide an indication of which available variables were most useful to the selected model.

14. Limitations and Future Improvements

• The prediction quality depends on the features available in the restaurant dataset and may not generalize to datasets with substantially different distributions.

• High-cardinality text fields such as Cuisines and Restaurant Name were excluded from the final model to keep the feature space practical; richer text or cuisine representations could be explored in future versions.

• The model is a regression system for estimating Aggregate rating and should not be interpreted as proving causal relationships between features and ratings.

• Future improvements could include hyperparameter search, ensemble regressors such as Random Forest or Gradient Boosting, additional feature engineering, and more systematic cross-validation.

15. Conclusion

A complete machine learning regression workflow was developed to predict restaurant Aggregate rating. The dataset was explored and preprocessed, missing cuisine values were handled, unsuitable and target-leaking features were removed, categorical variables were encoded, and the data was divided into training and testing sets. Linear Regression and Decision Tree Regression were evaluated, and the Decision Tree was further tuned to reduce overfitting. The final Decision Tree with max_depth=5 achieved strong test performance and provided feature importance information, demonstrating the usefulness of the selected machine learning approach.

16. Submission Readiness Checklist

Item	Status / Note
Task 1 notebook	Completed and tested from top to bottom
Source code	Included in the notebook
Model evaluation	MSE and R^2 reported for regression models
Model tuning	Decision Tree depths 5, 10, and 15 evaluated
Feature importance	Analyzed using the final Decision Tree
Visualization	Actual vs Predicted plot included
Documentation / report	This PDF report
Project demonstration video	Optional according to internship guidance
ZIP submission	Combine notebook, dataset if permitted/required, report, graph/output files, and any required model files into one ZIP